HDDA Industrial Track Proposal: Regularized Canonical Correlation Analysis for `statsmodels`: Estimation, Penalty Selection, and Significance Testing

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***Abstract*—This is an industrial-track project, direction "Linking two data sets". The deliverable is an extension of the `CanCorr` class in `statsmodels` that (i) keeps working when features outnumber samples ($p > n$) and (ii) states whether a canonical correlation is real or chance. The result will be submitted as a tested pull request with documentation and a worked example.**

I. Problem Statement

In one sentence: Python has no canonical correlation analysis (CCA) that works when features outnumber samples *and* provides significance testing. The current `statsmodels` implementation of CCA is a plain SVD-based procedure [1]: its fitting routine aborts with `ValueError("exog is collinear")` whenever a view is rank-deficient, which is always the case for $p > n$, and its only significance test, an asymptotic Wilks' Λ F-approximation, is derived under $n > p$. The scikit-learn implementation offers no regularization and no inference [2]; `cca-zoo` offers regularization but no significance testing [3]. We will implement:

- **Ridge-regularized fitting:** the whitening step of the SVD-based fit is replaced by a shrunken inverse, $1/s_i \to s_i/(s_i^2 + \lambda)$, which reduces exactly to the current implementation at $\lambda = 0$ [4], [5].
- **Penalty selection:** K-fold cross-validation maximizing the held-out canonical correlation, with all centering done inside folds to avoid leakage.
- **Inference:** a permutation test for each canonical correlation, reported next to the existing Wilks' Λ approximation.
- **Packaging:** tests, numpydoc documentation, and an example notebook, submitted as a pull request to `statsmodels/statsmodels`.

II. Evidence of Need

Community evidence comes from the `statsmodels` issue tracker itself. Issue #2625, opened in 2015 and still open on the wishlist, tracks reduced-rank regression—a family of models mathematically related to CCA—and its discussion collects the literature on penalized and high-dimensional variants, noting the connection to canonical correlation and rank testing [6]. Issue #4175 treats CCA as a supervised latent-variable method, gathers the rotation and reduced-rank literature around it, and links it to further missing tools [7]. Most directly, issue #1111 is titled "missing tools: canonical correlation and ranktest" [8]. All three issues have been planned or pending for years—without the functionality being delivered. Users also hit the failure in practice: for example, Stack Overflow question 75941931 reports `CanCorr` raising `ValueError: exog is collinear` on wide data [9].

Methodological evidence comes from stability studies of brain–behaviour CCA/PLS, which show that unregularized links fail to replicate at realistic sample sizes and recommend regularization together with resampling-based validation [10]. Finally, the ecosystem comparison shows a gap: in R the need is covered by three packages—`CCA` (ridge CCA for $p \gg n$ [5]), `PMA` (sparse CCA [11]), and `CCP` (significance tests [12])—while in Python the functionality is split and incomplete.

III. Existing Solutions and Limitations

Existing options are `statsmodels`' `CanCorr`, `sklearn.cross_decomposition` (`CCA`, `PLSCanonical`), `cca-zoo`'s `rCCA`, and the R packages listed above. The first fails hard on collinear data and offers only an asymptotic test; the second is prediction-oriented and exposes neither penalties nor tests; the third has penalties with cross-validation but no inference and no `statsmodels`-style results and summary output; the R packages have the features but only in R and fragmented across three packages. What is missing is one tested implementation inside a mainstream Python statistics library that combines ridge-regularized fitting for $p \gg n$, data-driven penalty choice, and a per-correlation significance statement.

IV. Data

We will use one public real dataset and a controlled synthetic generator (Table I). Nutrimouse is a public nutrigenomics dataset shipped with the R packages `CCA` and `mixOmics` [13]: $n = 40$ mice, one view of 120 gene expressions and one view of 21 hepatic lipid concentrations. Since $120 > 40$, the current `CanCorr` fails on it out of the box, which makes it a natural demonstration case. The synthetic generator is a latent-factor model with k planted canonical correlations of known strength; it provides ground truth for type-I/power calibration of the permutation test and for subspace-recovery checks.

Table I
Data Planned for the Project

Dataset	Obj.	Feat.	Regime	Role
Nutrimouse, gene view	40	120	$p > n$	real data, demo
Nutrimouse, lipid view	40	21	—	real data, second view
Synthetic, low-dim.	500	20+20	$n > p$	regression tests vs. current code
Synthetic, high-dim.	50, 200	100+100, 500+200	$p > n$	type-I/power, subspace recovery

V. Baselines for Comparison

- **Current `CanCorr`:** at zero penalty in the $n > p$ regime its output must be reproduced to machine precision; this is the backward-compatibility regression test.
- **scikit-learn `CCA` and `PLSCanonical`:** estimation quality and runtime at $p > n$.
- **`cca-zoo` `rCCA`:** penalty selection and estimation quality.
- **R references `CCA::cc` and `PMA::CCA`:** reference outputs computed once and committed as test fixtures; no runtime R dependency is introduced.

VI. Success Metrics

Correctness: canonical correlations match the R reference with relative error below 10^{-6} in both regimes; estimated subspaces agree up to sign and scaling, measured by principal angles.

Robustness: the $p > n$ path runs without exceptions, correlations stay in $[0, 1]$, and cross-validation selects penalties without leakage.

Inference: the permutation test controls type-I error within Monte Carlo error at $\alpha = 0.05$ under the null, and its power is reported against the signal-to-noise ratio; on the synthetic grid the regularized estimator must beat the unregularized one at small n.

Engineering: the test suite passes including new tests ($\geq$ 90% coverage of new lines), `ruff` is clean, docstrings follow numpydoc, the documentation builds without warnings, and an example notebook reproduces every reported number.

Delivery: a pull request opened against `statsmodels/statsmodels` plus a benchmark table comparing accuracy and runtime against all baselines.

VII. Risks and Fallback Plan

The main risk is a conventions mismatch: penalty parameterizations (absolute vs. trace-relative) and coefficient normalizations differ across packages, so coefficient-level agreement with R or `cca-zoo` may be unattainable even when the implementation is correct. We mitigate this by adopting the trace-relative parameterization of Gonzalez et al. [5], certifying on sign- and scale-invariant quantities (canonical correlations, principal angles), and documenting the conventions explicitly.

The second risk is that upstream review is not finished within the semester. The fallback is an installable fork with CI, tests, documentation, and the benchmark suite, so that the graded artifact does not depend on the merge.

VIII. Expected Output

The final output is a tested pull request extending `CanCorr` with a penalty parameter, a cross-validation helper for penalty selection, and a permutation option in the significance test, together with documentation and an example notebook on Nutrimouse. Accompanying it is a benchmark report, written as a short paper draft, comparing the implementation against scikit-learn, `cca-zoo`, and the R references on real and synthetic data. Sparse and kernel CCA are explicitly out of scope and remain stretch goals.

References

[1] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with python," in *Proceedings of the 9th Python in Science Conference*, vol. 57, Austin, TX, 2010, p. 61.

[2] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in python," *J. Mach. Learn. Res.*, vol. 12, no. null, p. 2825–2830, Nov. 2011.

[3] J. Chapman and H.-T. Wang, "Cca-zoo: A collection of regularized, deep learning based, kernel, and probabilistic cca methods in a scikit-learn style framework," *Journal of Open Source Software*, vol. 6, p. 3823, 12 2021.

[4] H. Vinod, "Canonical ridge and econometrics of joint production," *Journal of Econometrics*, vol. 4, no. 2, pp. 147–166, 1976. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0304407676900105

[5] P. Besse, I. González, S. Déjean, P. Martin, O. Gonçalves, and A. Baccini, "Human abc transporters put in context with the level of expression with molecular candidate efficiency in cancerous cellular lines," *Bulletin du cancer*, vol. 94, no. 10, pp. M–S, 2007.

[6] statsmodels developers, "ENH: Reduced rank vector models," GitHub issue #2625, statsmodels/statsmodels, 2015. [Online]. Available: https://github.com/statsmodels/statsmodels/issues/2625

[7] ——, "Canonical correlation as supervised PCA, Factor Analysis," GitHub issue #4175, statsmodels/statsmodels, 2017. [Online]. Available: https://github.com/statsmodels/statsmodels/issues/4175

[8] ——, "Missing tools: Canonical correlation and rank test," GitHub issue #1111, statsmodels/statsmodels, 2014. [Online]. Available: https://github.com/statsmodels/statsmodels/issues/1111

[9] Stack Overflow, "`statsmodels` `CanCorr` raises "exog is collinear"," Stack Overflow question 75941931, 2023. [Online]. Available: https://stackoverflow.com/questions/75941931

[10] M. Helmer, S. Warrington, A.-R. Mohammadi-Nejad, J. L. Ji, A. Howell, B. Rosand, A. Anticevic, S. Sotiropoulos, and J. Murray, "On the stability of canonical correlation analysis and partial least squares with application to brain-behavior associations," *Communications Biology*, vol. 7, 02 2024.

[11] D. M. Witten, R. Tibshirani, and T. Hastie, "A penalized matrix decomposition, with applications to sparse principal components and canonical correlation analysis," *Biostatistics*, vol. 10, no. 3, pp. 515–534, 2009. [Online]. Available: https://doi.org/10.1093/biostatistics/kxp008

[12] U. Menzel, *CCP: Significance Tests for Canonical Correlation Analysis (CCA)*, 2022, r package. [Online]. Available: https://CRAN.R-project.org/package=CCP

[13] P. G. P. Martin, H. Guillou, F. Lasserre, S. Déjean *et al.*, "Nutrimouse: Gene expression and lipid concentrations in a nutrigenomic study," Dataset shipped with the R packages `CCA` and `mixOmics`, 2007. [Online]. Available: https://CRAN.R-project.org/package=CCA